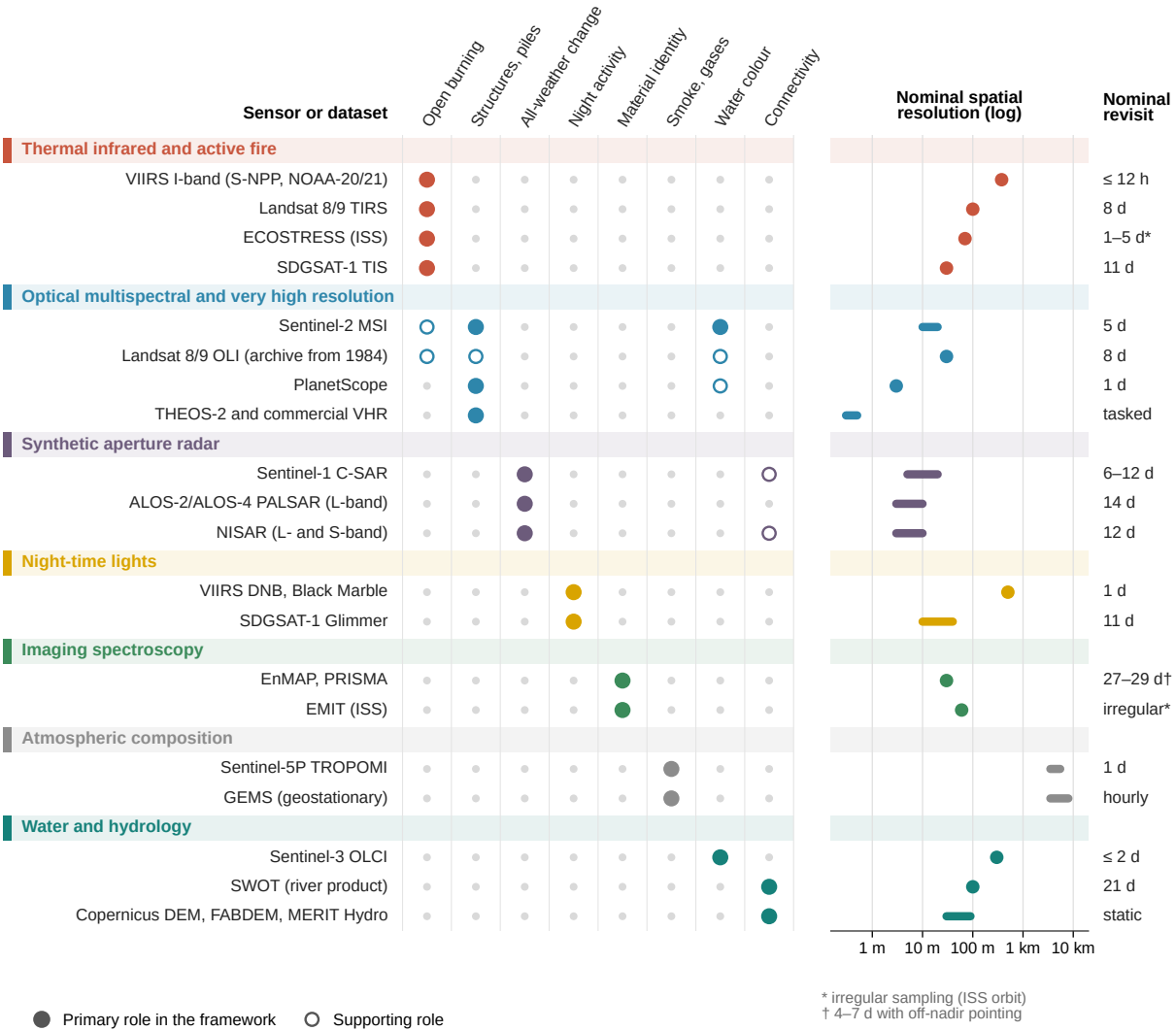
Sensor or dataset
Open burning
Structures, piles
All-weather change
Night activity
Material identity
Smoke, gases
Water colour
Connectivity
Nominal spatial resolution (log)
Nominal revisit
Thermal infrared and active fire
VIIRS I-band (S-NPP, NOAA-20/21)
≤ 12 h
Landsat 8/9 TIRS
8 d
ECOSTRESS (ISS)
1–5 d*
SDGSAT-1 TIS
11 d
Optical multispectral and very high resolution
Sentinel-2 MSI
5 d
Landsat 8/9 OLI (archive from 1984)
8 d
PlanetScope
1 d
THEOS-2 and commercial VHR
tasked
Synthetic aperture radar
Sentinel-1 C-SAR
6–12 d
ALOS-2/ALOS-4 PALSAR (L-band)
14 d
NISAR (L- and S-band)
12 d
Night-time lights
VIIRS DNB, Black Marble
1 d
SDGSAT-1 Glimmer
11 d
Imaging spectroscopy
EnMAP, PRISMA
27–29 d†
EMIT (ISS)
irregular*
Atmospheric composition
Sentinel-5P TROPOMI
1 d
GEMS (geostationary)
hourly
Water and hydrology
Sentinel-3 OLCI
≤ 2 d
SWOT (river product)
21 d
Copernicus DEM, FABDEM, MERIT Hydro
static
1 m
10 m
100 m
1 km
10 km
Primary role in the framework
Supporting role
* irregular sampling (ISS orbit)
† 4–7 d with off-nadir pointing